

RESEARCH ARTICLE

Sequential Deposition Strategy Enabling High-Performance Wide-Bandgap Perovskite Solar Cells With Fill Factor Exceeding 85%

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ABSTRACT

For wide-bandgap perovskite solar cells (WBG PSCs), significant non-radiative recombination at the self-assembled monolayers (SAM)/perovskite interface, along with high bulk defect densities, severely limit device performance and stability. Hence, defect suppression is critical to boost the fill factor (FF), representing a key challenge in developing high-performance devices. Herein, an innovative bilayer SAM (BL-SAM) interface engineering strategy is proposed via sequentially depositing [4-(3,6-dimethyl-9H-carbazol-9-yl) butyl] phosphonic acid (Me-4PACz) and (4-(3,11-dimethoxy-7H-dibenzo[*c,g*]carbazol-7-yl) butyl) phosphonic acid (MeO-4PADCB) as hole selective layers (HSLs). Theoretical and experimental evidence demonstrate that MeO-4PADCB not only passivates the defects caused by incomplete coverage at buried surface, but also interacts with the perovskite lattice. The BL-SAM strategy operates by orchestrating three synergistic effects: interfacial vacancy passivation, energy level alignment optimization, and residual lattice strain release. Consequently, the champion power conversion efficiency (PCE) of 20.35% is obtained with an exceptional FF of 85.18% and an open-circuit voltage (V_{OC}) of 1.33 V for 1.79 eV PSCs, representing one of the highest performance values reported in this category. The general applicability of this strategy is validated in 1.85 eV PSCs, delivering a PCE of 19.24% and an outstanding V_{OC} of 1.39 V. This work offers a versatile strategy for achieving high-performance and stable PSCs.

1 | Introduction

Tandem solar cells (TSCs) have garnered significant research interest in photovoltaics by virtue of their considerable potential to surpass the Shockley–Queisser efficiency limit through efficient solar spectrum utilization [1, 2]. Wide-bandgap per-

ovskite solar cells (WBG PSCs) ($E_g > 1.65$ eV), characterized by exceptional optoelectronic properties, serve as top sub-cells in TSCs for capturing high-energy short-wavelength photons [3–5]. However, WBG PSCs still face critical challenges, including significant non-radiative recombination losses and high bulk defect densities, particularly at the buried interface between the

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hole selective layers (HSLs) and the perovskite absorber [4, 6–10]. Self-assembled monolayers (SAM), leveraging their monomolecular thickness and tunable terminal functional groups, have demonstrated remarkable performance as hole-transport materials (HTMs) in WBG PSCs [1, 2, 11–15]. Nevertheless, conventional single component SAM often suffers from molecular aggregation, incomplete substrate coverage, and thermal instability, leading to current leakage and limited device performance [12, 16].

To address these inherent limitations of single SAM, recent research has increasingly focused on additive compensation and mixed SAM strategy [15, 17–19]. For instance, Fang et al. employed an interface engineering strategy based on the sequential deposition of [2-(9H-carbazol-9-yl) ethyl] phosphonic acid (Me-4PACz) and [4-(3,6-dimethyl-9H-carbazol-9-yl) butyl] phosphonic acid (2PACz). This approach enhanced hole extraction while simultaneously suppressed non-radiative recombination, enabling an optimized WBG device with a 1.78 eV bandgap to achieve a remarkable open-circuit voltage (V_{OC}) of 1.36 V and a record low V_{OC} deficit of 0.42 V, alongside a concurrently improved fill factor (FF) from 79.8% to 82.6% [20]. Li et al. proposed an in-situ compensation approach using strongly dipolar 5-aminopentanoic acid hydriodic acid (5-AVAI) molecules that fill vacancies in Me-4PACz during deposition, substantially reducing interfacial energy loss and achieving a high V_{OC} of 1.376 V in 1.79 eV WBG PSCs [21]. Annamaria et al. demonstrated that incorporating histamine molecules into Me-4PACz effectively modulates intermolecular interactions, thereby facilitating hole extraction and suppressing recombination losses. The collective improvements in interfacial energetics and film quality culminated in an enhanced FF of 83.38% for the 1.77 eV device. Consequently, this approach enabled inverted WBG PSCs with bandgaps of 1.77 and 1.83 eV to achieve champion power conversion efficiencies (PCE) of 20.34% and 18.99%, respectively [22]. Notably, a breakthrough interfacial modification strategy utilizing 6-aminohexane-1-sulfonic acid (SA) as a co-adsorbent with Me-4PACz was developed. This SA/Me-4PACz hybrid SAM effectively filled the uncovered sites on the substrate, broke the agglomeration of Me-4PACz, and optimized the energy level alignment at the buried interface of 1.77 eV-WBG PSCs, resulting in a certified PCE of 20.21% and a FF value of 84.84% [23]. In previous studies, despite continuous improvements in PCE, the FF value has generally remained below 85%, failing to achieve a substantial breakthrough. This limited improvement in FF arises from two factors, inadequate suppression of bulk defects, which causes non-radiative recombination during carrier transport, and suboptimal interfacial energy-level alignment, which impedes carrier extraction. These unresolved issues jointly constrain charge transport and hinder further enhancement of the FF.

To break through the bottleneck limiting FF enhancement, in this work, we proposed an innovative buried interfacial modification strategy employing the sequential deposition of Me-4PACz and (4-(3,11-dimethoxy-7H-dibenzo[c,g]carbazol-7-yl) butyl) phosphonic acid (MeO-4PADCB) as the HSLs. This bilayer SAM (BL-SAM) strategy simultaneously achieves synergistic effects at the buried interface, including interfacial vacancy passivation, energy level alignment optimization, and residual lattice strain release. As a result, we obtained a champion PCE

of 20.35% with an exceptional FF of 85.18%, a V_{OC} of 1.33 V, and a short-circuit current density (J_{SC}) of 17.94 mA·cm⁻² for 1.79 eV PSCs. Moreover, this strategy demonstrates broad applicability, enabling 1.85 eV devices to attain a PCE of 19.24% and a V_{OC} of 1.39 V. The BL-SAM devices also exhibit significantly enhanced operational stability, retaining 87% of their initial PCE after 400 h of continuous maximum power point (MPP) tracking under one-sun illumination.

2 | Result and Discussion

2.1 | Optoelectronic Performance Characterization of the Devices

The inherent amphiphilicity of SAM and its interactions in polar solvents can induce molecular aggregation and adsorption vacancy defects, which consequently compromise the optoelectronic performance and operational stability of the photovoltaic device [14]. To address this, we propose an innovative interface engineering strategy, namely BL-SAM strategy. This approach employs a sequential deposition process, in which Me-4PACz and MeO-4PADCB are spin-coated successively on the ITO conductive glass substrate (Figure S1). The strategy is designed to fill adsorption vacancies and passivate defects to stabilize the perovskite structure. Based on the BL-SAM strategy, inverted devices with the configuration of ITO/HSLs/Cs_{0.2}FA_{0.8}Pb(I_{0.6}Br_{0.4})₃/PEAI/PCBM/BCP/Ag were fabricated to investigate their photovoltaic properties (Figure 1a). The effective illumination area of each device was 0.06 cm². UV-vis absorption measurements confirmed that the BL-SAM strategy does not alter the perovskite bandgap (1.79 eV) (Figure S2). Through optimization of various parameters, including different SAMs, antisolvents and passivators (Figures S3–S7 and Tables S1–S5), the champion devices were ultimately obtained. The current density–voltage (J – V) curves of the champion devices are presented in Figure 1b. The control device based on single-layer Me-4PACz and MeO-4PADCB separately achieved the maximum champion PCE of 19.20% and 19.50%. In contrast, the BL-SAM device delivered a champion PCE of 20.35%, with an FF of 85.18%, a V_{OC} of 1.33 V, and a J_{SC} of 17.94 mA·cm⁻². Analysis of the data reveals that the enhancement in PCE primarily stems from improvements in both V_{OC} and FF, the corresponding photovoltaic parameters are presented in Table 1. Meanwhile, we also fabricated champion devices utilizing a reverse bilayer structure of MeO-4PADCB/Me-4PACz (R-BL-SAM), and a monolayer structure composed of a blend of Me-4PACz and MeO-4PADCB (co-SAM) and compared their optoelectronic performance parameters (Table S6). The R-BL-SAM and co-SAM devices exhibited champion PCE of 19.32% and 19.52%, respectively, both demonstrating a clear performance deficit compared to the BL-SAM device.

Furthermore, the forward and reverse scan characteristics (Figure 1c) of BL-SAM device display negligible hysteresis index as 2.15%, substantially less than the 3.45% for Me-4PACz and 3.12% for MeO-4PADCB, indicating excellent device stability and reproducibility [24]. The integrated external quantum efficiency (EQE) spectrum of the BL-SAM device (Figure 1d) revealed a calculated J_{SC} of 17.92 mA·cm⁻², closely matching the value

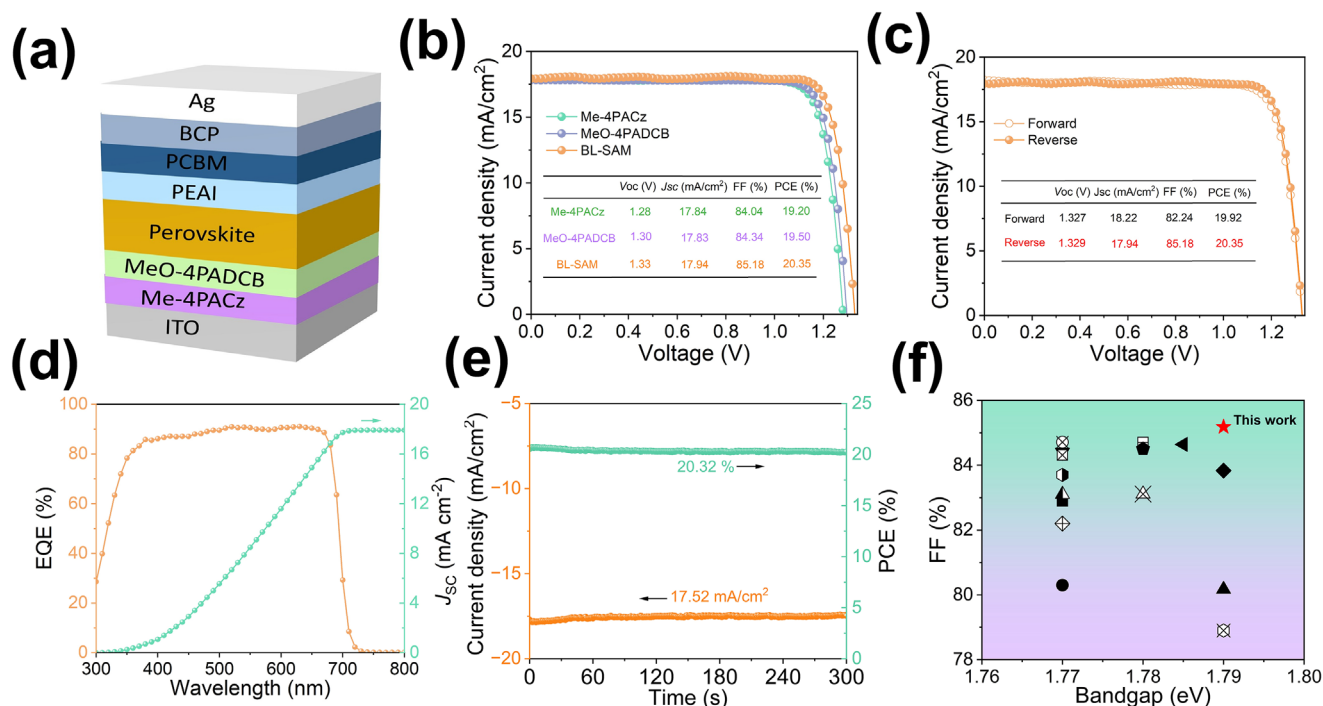


FIGURE 1 | (a) the schematic diagram of the inverted PSCs. (b) The champion J - V curves of inverted PSCs based on Me-4PACz, MeO-4PACz and BL-SAM devices. (c) J - V curves for the champion inverted PSCs based on BL-SAM devices. (d) EQE spectra and the integrated current density curves for the corresponding BL-SAM device. (e) The steady-state outputs of the devices with BL-SAM. (f) Reported FF vs. bandgap values for WBG PSCs, with the result of this work employing the BL-SAM strategy highlighted by the red star.

TABLE 1 | The photovoltaic parameters of BL-SAM, Me-4PACz, and MeO-4PACz champion devices.

		V_{OC} (V)	J_{SC} (mA·cm ⁻²)	FF (%)	PCE (%)	Hysteresis index (%)
BL-SAM	Reverse	1.329	17.94	85.18	20.35	2.15
	Forward	1.327	18.22	82.24	19.92	
Me-4PACz	Reverse	1.281	17.84	84.04	19.20	3.45
	Forward	1.280	17.90	80.99	18.56	
MeO-4PACz	Reverse	1.297	17.83	84.33	19.50	3.12
	Forward	1.296	17.89	81.51	18.91	

obtained from the J - V measurements. Additionally, as illustrated in Figure 1e, the BL-SAM device performed a stabilized power output (SPO) of 20.32% and a steady-state photocurrent of 17.52 mA·cm⁻², directly confirming its high reliability and practical performance. It is noteworthy that we achieved a champion FF of 85.18%, which stands as one of the highest FF values documented for 1.77–1.79 eV WBG PSCs (Figure 1f and Table S7). To assess the universal applicability and reproducibility of the BL-SAM strategy, we also fabricated 1.85 eV bandgap PSCs (Figures S8 and S9). Compared with the control devices, the BL-SAM device achieved optimal parameters with a PCE of 19.24%, FF of 82.88%, V_{OC} of 1.39 V, and J_{SC} of 16.68 mA·cm⁻². Moreover, as depicted in Figure S10, statistical analysis of 20 independent devices further confirmed the robustness and reproducibility of this interface engineering approach.

2.2 | Characterization and Measurements of the SAM Layer

The two SAM molecules, Me-4PACz and MeO-4PACz, are depicted in Figure 2a with their molecular structures. Notably, compared to Me-4PACz, MeO-4PACz possesses a larger molecular dipole moment and incorporates methoxy functional groups. To examine the adsorption differences between the two molecules on the ITO substrate, density functional theory (DFT) calculations were employed to quantify their binding energies. The adsorption energies (E_{ads}) for Me-4PACz and MeO-4PACz on the ITO substrate (Figure S11) were measured as −2.28 and −2.92 eV, respectively. This enhanced adsorption enables MeO-4PACz to effectively fill the adsorption vacancies caused by the localized aggregation of Me-4PACz molecules, leading to the

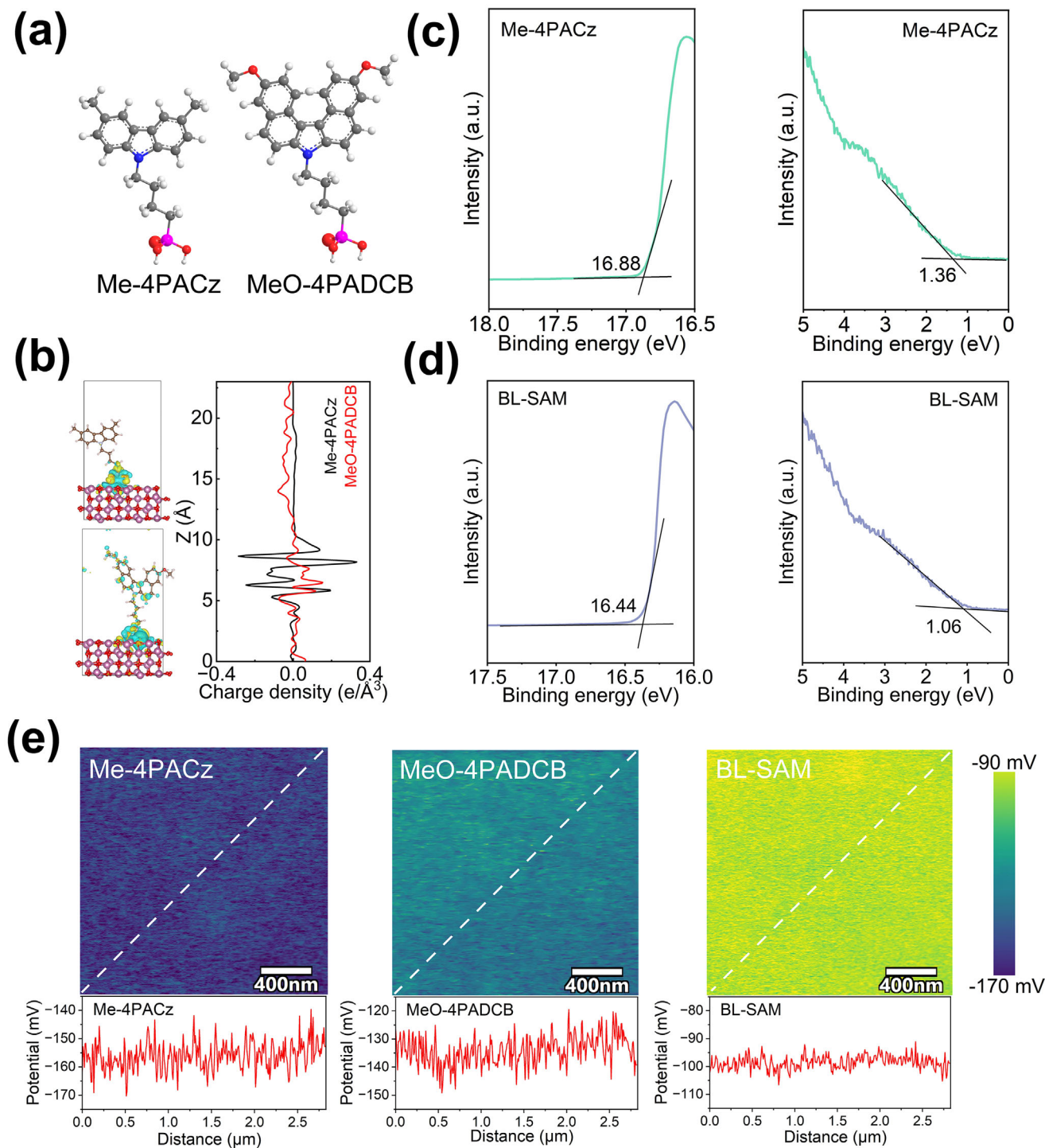


FIGURE 2 | (a) Molecular structures of Me-4PACz and MeO-4PADCBC. (b) The differential charge density of Me-4PACz and MeO-4PADCBC on the ITO substrate (left) and corresponding vertical linear distribution (right). (c,d) UPS diagrams for Me-4PACz and BL-SAM. (e) KPFM images and the corresponding contact potential difference (CPD) for Me-4PACz, MeO-4PADCBC and BL-SAM. The white dashed line corresponds to the distribution profile of CPD.

formation of a continuous and dense film coverage. Consequently, the defect density at the bottom interface is significantly reduced, providing a favorable foundation for the uniform growth of the subsequent perovskite layer [25]. Based on the differential charge density distribution analysis (Figure 2b), a wider charge transfer coverage was observed at the ITO/MeO-4PADCBC interface than ITO/Me-4PACz, indicating the enhancement of

charge transport and mitigation of interfacial recombination [26]. Contact angle measurements using perovskite precursor solution (Figure S12) revealed that the BL-SAM surface possessed the best wettability (19.7°), better than both Me-4PACz (32.5°) and MeO-4PADCBC (21.8°) monolayers. This superior wettability establishes a favorable interfacial foundation for the deposition of high-quality perovskite films.

Ultraviolet photoelectron spectroscopy (UPS) measurements were conducted to elucidate energy level impact of BL-SAM strategy. As shown in Figure 2c,d, the results demonstrate a distinct downward shift in both the work function (WF) and the highest occupied molecular orbital (HOMO) level for the BL-SAM. Specifically, BL-SAM exhibits a WF of -4.78 eV and a HOMO level of -5.84 eV, whereas Me-4PACz presented corresponding values of -4.34 and -5.70 eV, respectively. This favorable energy level alignment facilitates an improved p-type contact with the perovskite layer and results in enhanced hole extraction efficiency [25, 27]. While the WF and HOMO levels of the perovskite film were 4.30 and 5.92 eV, respectively (Figure S13). Utilizing these values, we constructed a comparative energy level diagram for Me-4PACz/perovskite and BL-SAM/perovskite. As shown in Figure S14, this visualization demonstrates that the BL-SAM architecture reduces the hole transport barrier by establishing a stepped energy level alignment. To further characterize the charge transport properties, the conductivity (σ) was calculated using the following formula [20, 27]:

$$\sigma = \frac{d}{AR} \quad (1)$$

where d is the SAM layer thickness, A is the active device area, and R is the resistance derived from J - V curves. As shown in Figure S15, the BL-SAM delivered significantly higher in-plane conductivity with 1.4×10^{-6} S/cm relative to both Me-4PACz (6.0×10^{-7} S/cm) and MeO-4PADCb (5.8×10^{-7} S/cm). This improved conductivity is attributed to the formation of a denser and more uniform molecular, which contributes to enhanced charge carrier mobility [28]. Furthermore, kelvin probe force microscopy (KPFM) characterization and the corresponding contact potential difference (CPD) analysis were utilized to evaluate interface homogeneity. As illustrated in Figure 2e, the markedly more uniform surface potential of BL-SAM than a single SAM layer, substantiates the formation of a highly homogeneous interface and an efficient non-radiative hole extraction pathway, which serves as a fundamental requirement for the attainment of a high FF [29–31].

2.3 | Characterization and Measurements of the Perovskite Films

The achievement of high-performance devices is fundamentally governed by the overall quality of the perovskite thin film, as characterized by crystalline quality, defect density and degree of non-radiative recombination [6–10]. To evaluate the underlying mechanisms of the enhanced film quality, DFT calculation was further adopted to validate the interaction between the BL-SAM structure and the perovskite layer. As depicted in Figure 3a, the E_{ads} of MeO-4PADCb on FAPbI₃ (011) crystal plane reaches -2.26 eV, considerably surpassing that of Me-4PACz with -0.85 eV. This enhanced adsorption reflects stronger coordinative interaction between the terminal functional groups of MeO-4PADCb and the undercoordinated Pb²⁺ ions in the perovskite layer, thereby effectively supporting lattice stabilization [27]. Consistent with the theoretical predictions, X-ray photoelectron spectroscopy (XPS) analysis reveals pronounced positive binding energy shifts in Pb 4f, I 3d, and Br 3d spectra for the BL-SAM film (Figure 3b,c, and Figure S16), along with slight shifts in the P—OH

and P=O groups of the O 1s spectrum (Figure S17). Simultaneously, Fourier-transform infrared spectroscopy (FTIR) analysis revealed that the C—O group peak underwent a redshift from 1237 cm⁻¹ in monolayer MeO-4PADCb to 1234 cm⁻¹ in the mixture of MeO-4PADCb with PbI₂ (Figure S18). These results confirm the interaction between the oxygen atoms of MeO-4PADCb and Pb²⁺ ions within the perovskite lattice, enabling effective passivation of interfacial defects [20, 32].

The improved crystallization behavior induced by the BL-SAM strategy is consistently reflected across multiple morphological characterizations. X-ray diffraction (XRD) analysis reveals a pronounced enhancement in the crystallographic quality of the BL-SAM film (Figure 3d), as evidenced by the significant augmentation of (100) diffraction peak intensity compared to the control samples. Notably, the (100)/(110) peak area and intensity ratios reached 0.804 and 0.726, respectively, substantially exceeding those of the single-layer Me-4PACz and MeO-4PADCb films (Figure S19). This explicitly demonstrates that the BL-SAM architecture promotes a preferential (100) orientation at the buried interface, which correlates with enhanced crystallographic alignment and reduced defect density [33]. Correspondingly, scanning electron microscopy (SEM) observations (Figure S20) and grain size distribution statistics (Figure 3e) demonstrate that the BL-SAM film possesses significantly enlarged grain dimensions and a lower density of defects [34]. This improvement in film quality is further corroborated by atomic force microscopy (AFM) of the buried interface. As illustrated in Figure S21, the average roughness of the BL-SAM film is reduced to 0.793 nm, in contrast to 1.895 nm for Me-4PACz and 2.089 nm for MeO-4PADCb, reflecting an obviously smoother interface associated with superior crystallinity. Collectively, the introduction of the BL-SAM structure modulates the crystallization behavior of the perovskite films, enabling more orderly crystal growth while simultaneously suppressing defect intensity. This dual effect thereby enhances their crystalline quality and structural integrity [35–37].

To elucidate the microstructural regulatory mechanism of the BL-SAM strategy on the buried surface, grazing incidence wide angle X-ray scattering (GIWAXS) measurements were performed. It should be clarified that the X-ray beam penetrated the perovskite layer to reach the ITO substrate at the diffraction angle of 0.8° (Figure S22) [35, 36]. When the incident angle was increased from 0.2° to 0.8°, as presented in Figure 3f and Figure S23, the characteristic diffraction peak of the BL-SAM film at $q = 2.29 \text{ \AA}^{-1}$ remained essentially stable, whereas pronounced peak shifts toward lower q values were observed for the control samples, which may be attributed to lattice distortion induced by vacancy defects within the film. The more stable trend of q value variation indicates that the BL-SAM strategy effectively mitigates residual strain at the buried interface of the perovskite film [38].

To further elucidate the kinetic influence of the BL-SAM strategy on perovskite crystallization, in situ photoluminescence (PL) measurements were conducted during spin-coating (Figure S24). Following antisolvent addition, the PL intensity of the Me-4PACz film peaked rapidly, whereas the BL-SAM film showed a notably delayed maximum [39–41]. This distinct kinetic behavior indicates that the BL-SAM structure effectively modulates

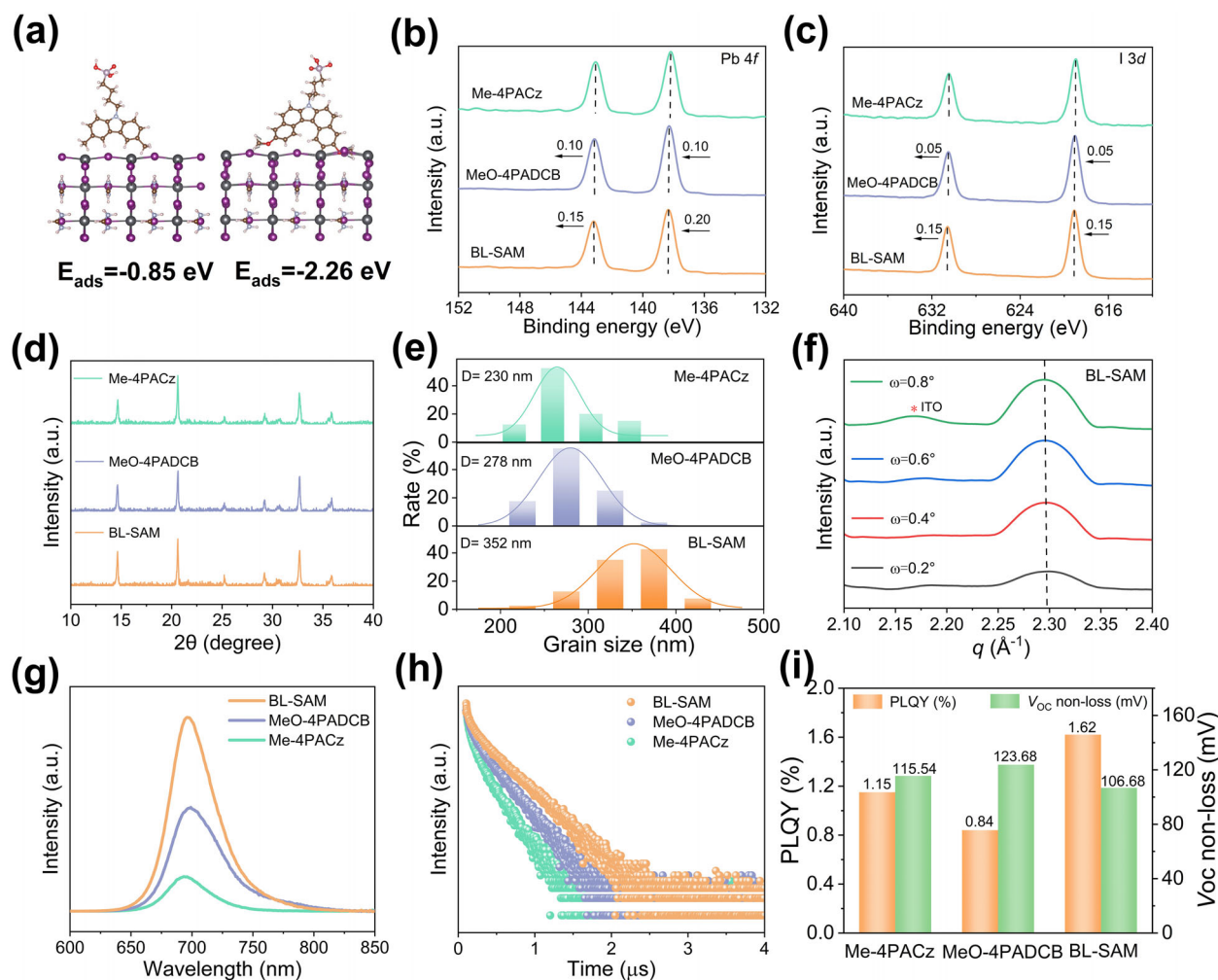


FIGURE 3 | (a) The E_{ads} of Me-4PACz and MeO-4PADCB on the FAPbI_3 (011) crystal plane. (b,c) XPS spectra of Pb 4f and I 3d in the perovskite buried interface for different SAMs. (d) XRD patterns of perovskite films with different SAMs. (e) the histograms of grain size distributions for the perovskite films. (f) GIWAXS patterns of BL-SAM perovskite films. (g,h) PL and TRPL spectra of perovskite films deposited on the bare glass. (i) PLQY and the corresponding V_{OC} non-radiative recombination loss.

the crystallization pathway, favoring a more ordered perovskite lattice, consistent with the larger grains and enhanced orientation observed in SEM and XRD. Meanwhile, we further investigated the non-radiative recombination dynamics within the perovskite films via photoluminescence (PL) and time-resolved photoluminescence (TRPL) analyses (Figure 3g,h). In comparison to the control samples, the BL-SAM film exhibits higher PL intensity and longer carrier lifetime, demonstrating efficient suppression of non-radiative recombination, which plays a critical role in minimizing interfacial carrier losses and achieving an excellent FF value [42–45]. Consistent with the aforementioned results, the calculated non-radiative open-circuit voltage loss (ΔV_{OC}) (Figure 3i), derived from photoluminescence quantum yield (PLQY) via the following formula [20, 46–48]:

$$\Delta V_{\text{OC}} = -K_{\text{B}} T \ln(\text{PLQY}) / q \quad (2)$$

where K_{B} denotes the Boltzmann constant, T stands for the temperature, and q is the elementary charge. The lowest ΔV_{OC} value of 106.68 mV for the BL-SAM film is obtained corresponding to the highest PLQY value with 1.62%, providing clear evidence

of effectively reducing voltage losses [20, 23]. Overall, the suppressed voltage loss confirms the effectiveness of the BL-SAM strategy in minimizing interfacial non-radiative recombination, coupled with the improvement of the perovskite film quality, which underpins the achievement of devices with outstanding PCE and FF.

2.4 | Regulation Mechanism of the BL-SAM Strategy

Based on the aforementioned conclusions, we deduce that the BL-SAM strategy enables the simultaneous realization of synergistic effects at the buried interface, including passivation of vacancy defects, optimization of energy level alignment, and release of lattice strain. As depicted in Figure 4, based on DFT-optimized geometries and molecular spacing and alignment relative to the perovskite lattice, a schematic diagram depicting the mechanism of the BL-SAM strategy has been prepared [49]. In the case of the single-layer Me-4PACz, intermolecular aggregation leads to adsorption vacancies on the ITO substrate,

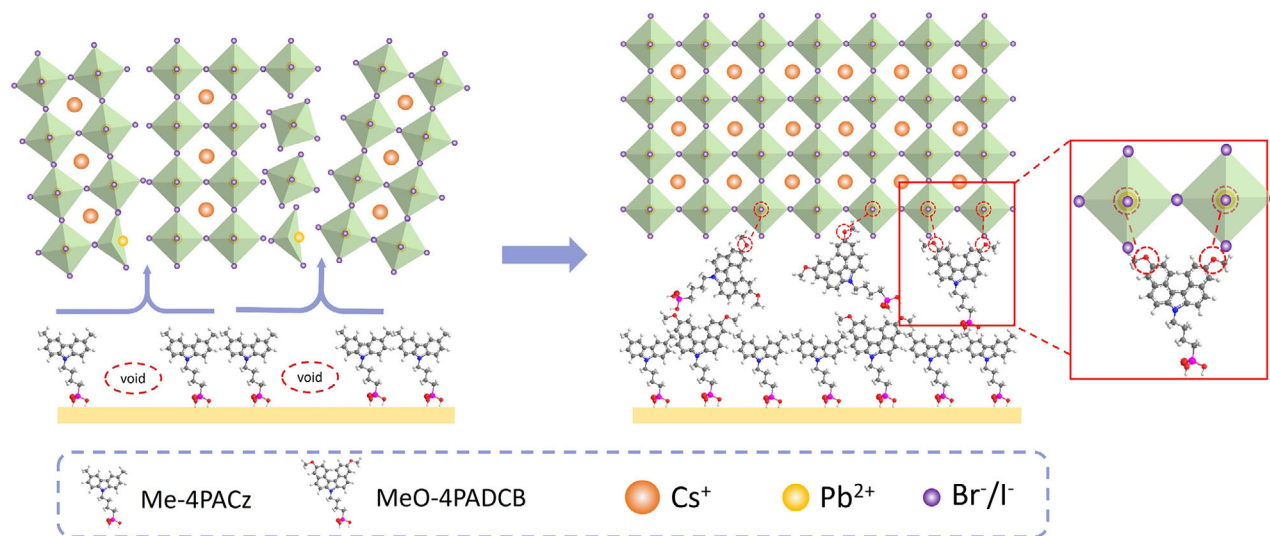


FIGURE 4 | Schematic diagram of the BL-SAM interface engineering strategy mechanism. As the second layer of HSL, MeO-4PACB fills the bottom interface adsorption vacancy caused by Me-4PACz, regulating the deposition and crystallization process at the bottom of perovskite.

resulting in partial substrate exposure. This subsequently induces lattice distortion and residual stress accumulation at the buried perovskite interface, ultimately causing interfacial structural collapse. Such degradation significantly increases defect density, thereby limiting the overall optoelectronic performance and long-term stability of the device.

In contrast, the BL-SAM interface engineering strategy yields pronounced optimization. The underlying Me-4PACz, functionalized with phosphate groups, forms a stable anchor with the ITO substrate, providing robust structural support to the interface. The upper MeO-4PACB layer, possessing stronger adsorption capability, not only fills the surface vacancies caused by Me-4PACz aggregation, thus enhancing the continuity and compactness of the molecular layer, but also coordinates with undercoordinated Pb^{2+} ions in the perovskite lattice via its terminal methoxy groups. This effectively suppresses ion migration and stabilizes the crystal lattice, substantially alleviating interfacial residual stress and reducing interfacial defect density.

Such synergistic interaction not only reinforces the structural stability and energy level alignment at the interface but also modulates film crystallization and optimizes interfacial charge carrier transport. Consequently, the overall device performance is substantially elevated, largely driven by the pronounced improvement in FF.

2.5 | Carrier Dynamics and Stability Characterization of the Devices

To assess comprehensive assessment of the carrier dynamics and defect passivation, the dependence of V_{OC} on light intensity was investigated, as depicted in Figure 5a. The relationship between V_{OC} and light intensity can be described by the formula [39, 40, 50]:

$$V_{\text{oc}} = \frac{nK_{\text{B}}T}{q} \ln(I) + A \quad (3)$$

where n , K_{B} , T , q , I and A represent the diode ideality factor, Boltzmann constant, temperature in Kelvin, elementary charge, light intensity and a constant, respectively. The BL-SAM device exhibited an n of 1.60, which is lower than the values of 1.95 and 1.74 measured for Me-4PACz and MeO-4PACB devices, respectively. The lower n value indicates diminished trap-assisted recombination, indicating effective suppression of Shockley-Read-Hall recombination pathways [51, 52]. Furthermore, we evaluated the influence of the BL-SAM strategy on the defect state density (N_{trap}) through space-charge-limited current (SCLC) measurements. As shown in Figure 5b, the trap-filled limit voltage (V_{TFL}) is defined as the voltage at the transition point between the ohmic and trap-filled regions. The N_{trap} is obtained via the following formula [41, 53]:

$$N_{\text{trap}} = \frac{2\epsilon_r\epsilon_0 V_{\text{TFL}}}{qL^2} \quad (4)$$

where ϵ_r represents the relative dielectric constant, ϵ_0 denotes the vacuum dielectric constant, q stands for the elementary charge, and L corresponds to the thickness of the perovskite film. The BL-SAM device achieved the lowest V_{TFL} , at only 0.17 V, which also signifies that it has the lowest N_{trap} [52, 54], indicating that the defect density is distinctly reduced within the PSCs [55, 56]. To further analyze the charge recombination behavior within the devices, we performed transient photovoltage (TPV) (Figure 5c) and transient photocurrent (TPC) (Figure S25) measurements. A longer carrier lifetime indicates lower recombination losses, while a shorter charge extraction time suggests higher charge collection efficiency at the electrodes [57–60]. Notably, the prolonged carrier lifetime (4.72 μs) and shortened extraction time (1.79 μs) observed in the BL-SAM device confirm suppressed internal recombination and enhanced charge transport efficiency [59–61]. Thus, these favorable kinetic parameters constitute the mechanistic basis for the superior FF value [42].

Meanwhile, we systematically conducted long-term stability assessments on unencapsulated PSCs under various conditions to comprehensively evaluate the effectiveness of the BL-SAM

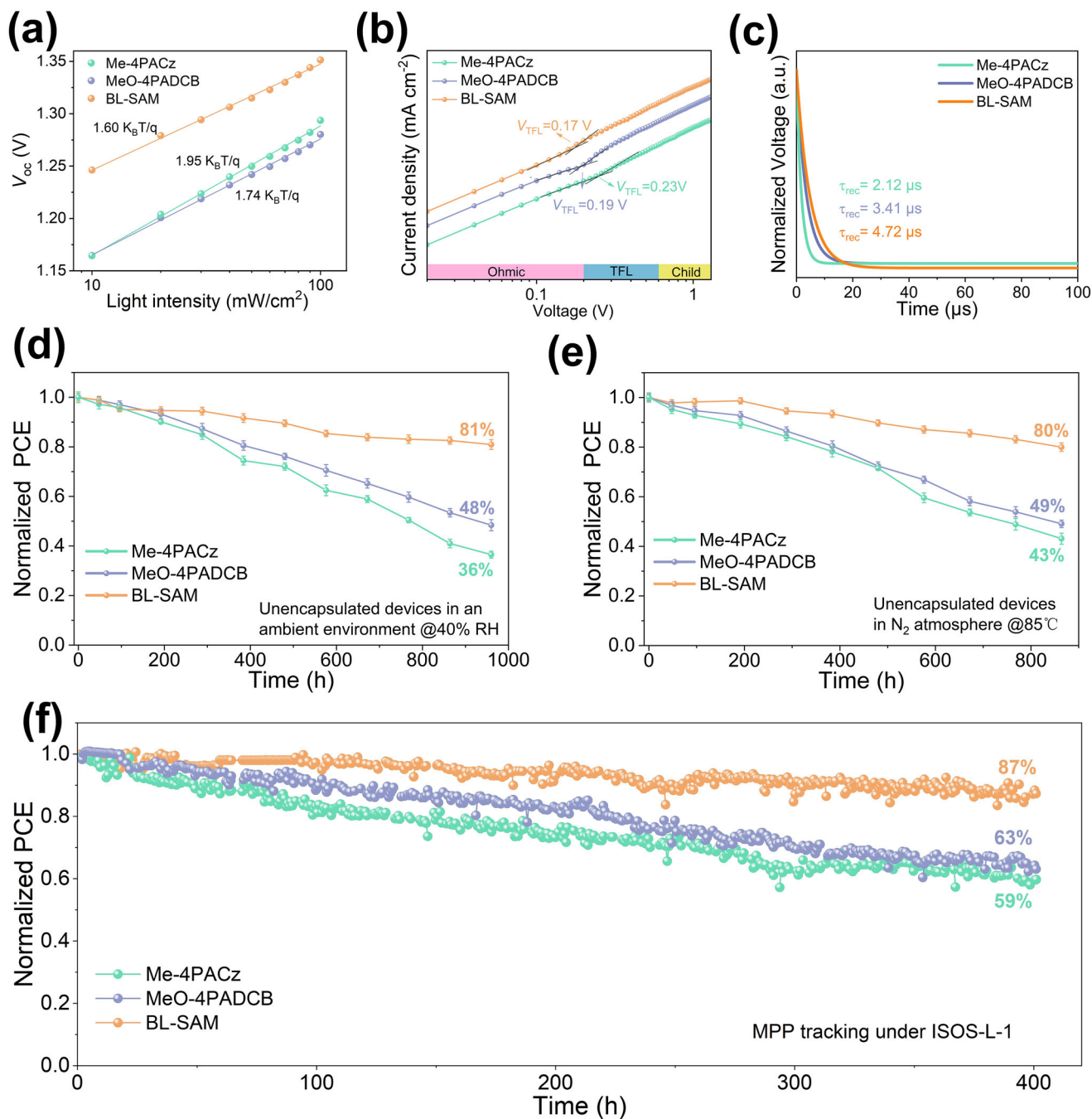


FIGURE 5 | (a) Schematic diagram of V_{OC} light intensity dependence of PSCs. (b) SCLC diagram-based hole-only devices with the structure of ITO/SAM/perovskite layer/ MoO_3/Ag . (c) TPV of the PSCs. Long-term stability tests of unpackaged devices: (d) stability test at 40% relative humidity at room temperature ($\sim 25^\circ\text{C}$) in ambient environment; (e) thermal stability test at 85°C temperature in N_2 atmosphere; (f) maximum power point (MPP) tracking stability test in continuous light environment within the temperature range of $30^\circ\text{C} \pm 5^\circ\text{C}$.

strategy in enhancing device operational stability. As illustrated in Figure 5d, humidity stability was evaluated by exposing unencapsulated devices under ambient environmental conditions (25°C and RH of 40%). After 960 h of exposure, the BL-SAM modified device maintained 81% of its initial PCE, in contrast to the control devices, which degraded to 48% and 36%, respectively. As shown in Figure 5e, thermal stability was examined under an inert nitrogen atmosphere at an elevated temperature of 85°C . After 864 h of thermal aging, the BL-SAM devices retained approximately 80% of their initial PCE, while the control counterparts exhibited

substantial degradation, maintaining only 49% and 43% of their original efficiencies, respectively.

Additionally, according to the ISOS-L-1 protocol, MPP tracking stability measurements were carried out under continuous one-sun illumination at $30 \pm 5^\circ\text{C}$, as presented in Figure 5f. After 400 h of continuous light exposure, the BL-SAM device retained 87% of its initial performance, whereas the efficiencies of the reference devices decreased to 63% and 59%. The results demonstrate that the BL-SAM interface regulation strategy significantly enhances

thermal, moisture, and light stability of the devices. This is primarily attributed to its effective suppression of non-radiative recombination at the interfaces, control over the perovskite crystallization processes and formation of dense films. Ultimately, the BL-SAM interface strategy achieves the synergistic effect of passivating interface defects and optimizing interface energy level matching, resulting in devices with high FF and potential for large-area preparation.

3 | Conclusions

In summary, we have fabricated high-performance and stable WBG PSCs based on the BL-SAM interface engineering strategy via sequentially depositing Me-4PACz and MeO-4PADCB. In contrast to conventional approaches such as molecular modulation and additive engineering, the BL-SAM strategy enables the triple synergistic effect. Specifically, it effectively fills the adsorption vacancies at the bottom interface caused by incomplete coverage of the monolayer SAM, optimizes the energy level alignment between the HSL and perovskite and modulates the crystallization process of the perovskite layer to release residual strain induced by excessive lattice contraction. As a result, this approach significantly enhances hole extraction efficiency while suppressing non-radiative recombination within the device. Consequently, the BL-SAM device achieved a champion PCE of 20.35% with an exceptional FF of 85.18%, representing state-of-the-art performance for 1.79 eV WBG PSCs. Additionally, this strategy proved to be equally applicable to devices with a bandgap of 1.85 eV, yielding a champion PCE of 19.24% with V_{OC} of 1.39 V. This research offers valuable insights that pave the way for developing high-performance tandem devices in the future.

4 | Experimental Section

4.1 | Materials

[4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl] phosphonic Acid (Me-4PACz) and [4-(7H-Dibenzo[c,g]carbazol-7-yl)butyl] phosphonic Acid (MeO-4PADCB) were purchased from Advanced Election Technology Co., Ltd. Cesium iodide (CsI, 99.99%), lead bromide ($PbBr_2$, 99.99%), methylammonium chloride (MACl), Methylammonium bromide (MABr) and bathocuproine (BCP), were purchased from Xi'an Polymer Light Technology Corporation. Lead iodide (PbI_2) was purchased from TCI. Formamidinium iodide (FAI, 99.99%), and propane-1,3-diaminium Iodide (PEAI) were purchased from Great Cell Solar. [6,6]-phenyl-C61-butyric acid methyl ester ($PC_{61}BM$), N,N-dimethylformamide (DMF), dimethylsulfoxide (DMSO), ethanol, Chlorobenzene (CB) and isopropanol (IPA) were purchased from Sigma-Aldrich.

4.2 | Fabrication Process of Inverted PSCs

4.2.1 | 1.79 eV PSCs

The perovskite precursor solution with a total concentration of 1.2 M was formulated by dissolving CsI, $PbBr_2$, PbI_2 , and FAI in a solvent mixture of DMF and DMSO with a ratio of 4:1. The molar

ratios of FA^+ to Cs^+ and I^- to Br^- were controlled at 8:2 and 6:4, respectively. To modulate the crystallization dynamics, 10 mol% of MACl was introduced into the precursor solution. The solution was stirred at 50°C for 2 h to ensure complete dissolution and homogenization, followed by cooling to room temperature prior to use. The ITO glass was cleaned successively with detergent, deionized water, acetone and ethanol, and ultrasonic treatment time was 15 min for each step. After cleaning, the ITO glass surface was dried with nitrogen, and then the ITO substrate was treated with ultraviolet ozone for 15 min. For the control devices, single-layer solutions of Me-4PACz (0.3 mg/mL in isopropanol) and MeO-4PADCB (0.3 mg/mL in isopropanol) were respectively spin-coated onto ITO glass substrates at 4000 rpm for 30 s, followed by thermal annealing at 100°C for 10 min under a nitrogen atmosphere and subsequently cooled to room temperature. For the target devices, the Me-4PACz (0.3 mg/mL in isopropanol) solution was first spin-coated and thermally annealed under the same conditions, then cooled to room temperature. Subsequently, the MeO-4PADCB (0.3 mg/mL in isopropanol) solution was spin-coated on top at 4000 rpm for 30 s, followed by a second annealing step at 100°C for 10 min and finally cooling to room temperature. Subsequently, perovskite films were deposited onto the bilayer SAM-modified substrate and spin-coated at 4000 rpm for 40 s. At 8 s before the end of the spin-coating process, 160 μ L of chlorobenzene (CB) was dropped as an antisolvent. The resulting films were transferred to a hot plate and annealed at 100°C for 15 min under a nitrogen atmosphere, followed by cooling to room temperature. Then, PEAi solution (1.5 mg/mL in IPA) was dynamically spin-coated at 4000 rpm for 30 s on the top of perovskite film, following by annealing at 100°C for 10 min and cooling to room temperature. The $PC_{61}BM$ solution (20 mg/mL in CB) was spin-coated (1500 rpm for 30 s) on top of the PEAi layer. Subsequently, BCP (0.5 mg/mL in IPA) was spin-coated on the $PC_{61}BM$ layer at 4000 rpm for 30 s. Then, a 100 nm thick silver (Ag) electrode was thermally evaporated onto the surface of the BCP layer under a high vacuum of 1.8×10^{-4} Pa. Ultimately, as an anti-reflective coating, a 100 nm thick magnesium fluoride (MgF_2) layer was thermally evaporated onto the rear side of the devices.

4.2.2 | 1.85 eV PSCs

The perovskite precursor solution with a total concentration of 1.0 M was formulated by dissolving CsI, $PbBr_2$, PbI_2 , MABr, and FAI in a solvent mixture of DMF and DMSO with a ratio of 4:1 with the chemical formula of $(FA_{0.8}Cs_{0.2})_{0.3}(FA_{0.6}MA_{0.4})_{0.7}Pb(I_{0.5}Br_{0.5})_3$. The solution was stirred at 50°C for 2 h to ensure complete dissolution and homogenization, followed by cooling to room temperature prior to use. The perovskite films were deposited onto the bilayer SAM-modified substrate by using a two-step spin-coating process (2000 rpm for 20 s and 4000 rpm for 15 s). At 8 s before the end of the spin-coating process, 160 μ L of CB was dropped as an antisolvent. The substrates were then transferred to a hotplate and heated at 105°C for 15 min under a nitrogen atmosphere, followed by cooling to room temperature. Except for the perovskite absorber layer, the solution concentrations, spin-coating parameters, and annealing conditions for the remaining layers were identical to those used in the PSCs of a 1.79 eV bandgap.

4.3 | Characterization and Measurement

4.3.1 | Film Characterization

Films with hole selective layer and perovskite film layer were fabricated on ITO glass substrate, Me-4PACz/PVK deposited ITO glass (denoted as “Me-4PACz”), MeO-4PACz/PVK deposited ITO glass (denoted as “MeO-4PACz”) and Me-4PACz / MeO-4PACz/PVK deposited ITO glass (denoted as “BL-SAM”) for subsequent characterization and comparative analysis. The bottom interface sample of the perovskite thin film was prepared by detaching the perovskite film from its ITO glass substrate using ultraviolet (UV) curing adhesive. Scanning electron microscopy (SEM) images were acquired using a NOVA NANOSEM 450 (Thermo Scientific, USA). Surface topography in the form of height images was characterized by atomic force microscopy (AFM) on an OXFORD MFP-3D system. X-ray photoelectron spectroscopy (XPS) measurements were performed using an X-ray photoelectron spectrometer (ESCALAB Xi+). Photoluminescence (PL) and time-resolved photoluminescence (TRPL) spectra were collected using an Edinburgh FLS-1000 instrument. The absorption characteristics of the samples were measured with a UV-vis spectrometer (Lambda 750, PerkinElmer). Grazing Incidence Wide Angle X-Ray Scattering (GIWAXS) measurements were conducted at the Suzhou Vacuum Interconnected Nanotech Workstation using X-ray radiation with a wavelength of approximately $\lambda \approx 1.54 \text{ \AA}$. The 2D GIWAXS patterns were obtained using a vertically mounted Mar CCD detector placed $\sim 194 \text{ mm}$ from the sample, under a grazing incidence angle of 0.5° , with an exposure time of 50 s. Data analysis was performed using FIT2D software, and the patterns were presented in terms of the scattering vector q , calculated by $q = 4\pi \sin\theta/\lambda$, here θ is half the scattering angle and λ denotes the incident X-ray wavelength.

4.3.2 | Devices Characterization

The current density-voltage (J - V) characteristics of the photovoltaic devices were measured at room temperature within a nitrogen-filled glovebox, employing a xenon lamp solar simulator (Enlitech, SS-X50, Taiwan). The illumination intensity was calibrated to 100 mW cm^{-2} using a silicon reference cell equipped with a KG2 filter. Device performance was evaluated using an IVS-KA6000 source meter under both reverse (1.35 to -0.1 V) and forward (-0.1 to 1.35 V) scanning modes, with a scan rate of 0.01 V s^{-1} and a delay time of 10 ms. The active area for all devices was defined as 0.06 cm^2 by a metal mask, and no preconditioning procedures were applied prior to testing. External quantum efficiency (EQE) measurements were performed in ambient conditions using a QE-R EQE system (Enlitech, Taiwan). Operational stability assessments were conducted using a multi-channel solar cell performance decay testing platform (PVL-T-G8001M, Suzhou D&R Instruments Co. Ltd.) inside a nitrogen-filled glovebox, where the levels of H_2O and O_2 were maintained at approximately 100 ppm and 80 ppm, respectively. Devices were subjected to continuous illumination provided by a white LED light source (D&R Light, L-W5300KA-150, Suzhou D&R Instruments Co. Ltd.) simulating one sun intensity, with the initial J_{SC} matching that under standard test conditions. Device performance was periodically tracked by I - V sweeps ranging from -1.35 to 0.2 V in 0.01 V increments. During the intervals between

sweeps, an external load resistor-matched to the maximum power point ($R = I_{\text{max}} / V_{\text{max}}$) was connected to ensure that the recorded PCE decay accurately reflected real-time performance degradation under simulated operating conditions. The MPP was automatically extracted based on the J - V data collected.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data are available from the corresponding author upon reasonable request.

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